

Advancing Legal Reasoning in *Algerian Law*

RAG · KNOWLEDGE GRAPHS · ARGUMENT MINING → Citation-faithful legal answers

Ibrahim El Khalil Attia · Maab Chaoui | Supervisor Dr. Fouad Dahak Co-supervisor Prof. Ivan Cantador

01 The Problem

WHY WHY LEGAL QA IS HARD

1.1 Fluent is not lawful

A legal answer cannot be judged by fluency or plausibility. To be valid it must be **grounded in identifiable sources**, cite the **exact governing articles**, stay inside the **Algerian jurisdiction**, and read each provision **as amended** at the relevant date — and where the law is silent, the only correct answer is a **refusal**. A fluent language model guarantees none of this.

SOURCED · ARTICLE-CITED · JURISDICTION-BOUND · AMENDMENT-AWARE · OR *REFUSED*

1.2 What legal questions demand

بين الأساس القانوني لمبدأ خضوع القاضي للنص التشريعي، وحدود سلطته عند غياب النص...

— demands exact articles — *Civil Code, arts. 1–2*

كيف تغيّرت شروط الخلع في قانون الأسرة بعد تعديل 2005؟

— demands the law as amended — *Ord. 05-02 → Family Code, art. 54*

هل يحق لصاحب العمل في الجزائر فصل العامل؟ (at-will) تامة دون إبداء أسباب

— a foreign-doctrine trap — *Algerian law is silent; the system must abstain*

verbatim from *AlgerianLegalBench* (244 questions)

1.3 Where today's LLMs fail

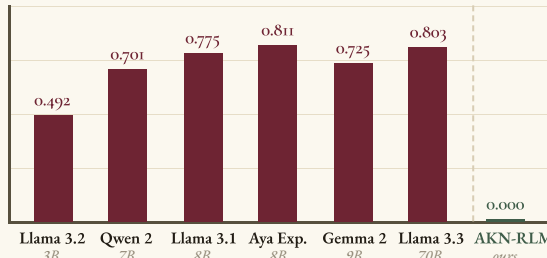


Fig. 1. Hallucinated-citation rate: the share of cited articles that *do not exist*, per direct-LLM baseline (244 questions). Scale does not help — the 70B model fabricates 4 of 5 citations. Gemini 2.5 reaches 0.008 only by declining to cite at all, and six of seven baselines import foreign law on >12 of 40 trap questions. AKN-RLM: 0.000.

1.4 ...and Algerian law raises the bar

- Arabic-first. 232 of 244 benchmark questions are in Arabic
- Layered in time. 1963–2025; amendments rewrite articles in place
- Deeply structured. code → chapter → section → article
- Low-resource. barely covered by existing legal-AI corpora

1.5 Objectives

- ✓ Structure the law
171 docs → Akoma Ntoso XML + legal KG
- ✓ Benchmark it
AlgerianLegalBench: 244 expert questions
- ✓ Reason on evidence
typed retrieval + bounded recursion
- ✓ Verify or abstain
triple faithfulness gate

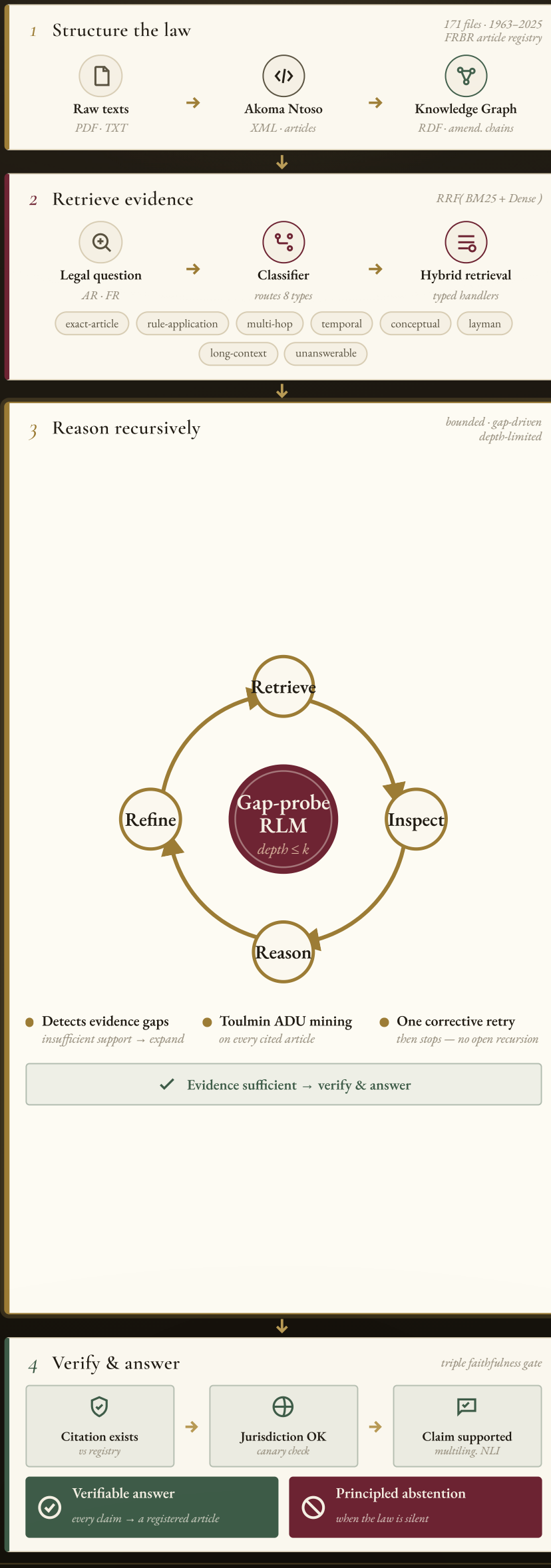
1.6 Significance & impact

- First citation-faithful QA system for Algerian law — every claim resolves to a registered article, or the system abstains.
- Reusable infrastructure for Algerian legal AI
 - AKN corpus + knowledge graph — 171 structured documents, 1963–2025, article-level registry
 - AlgerianLegalBench* — 244 expert-annotated questions, $\kappa = 0.829$
- A recipe for low-resource legal AI — evidence control beats parameter scale.

02 The AKN-RLM Architecture

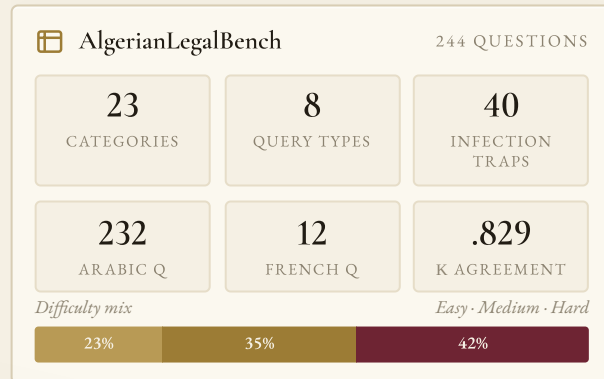
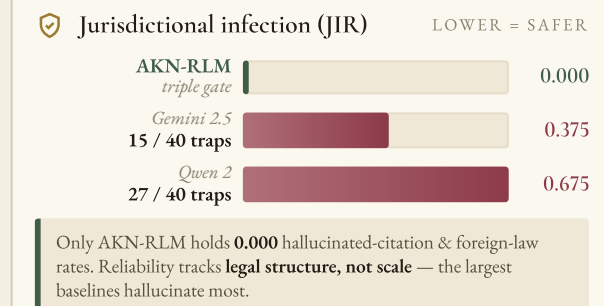
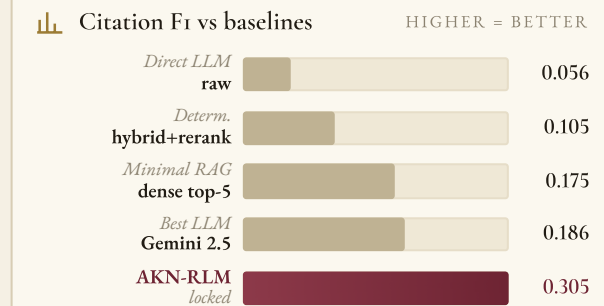
HOW DOCUMENTS → STRUCTURE → VERIFIED ANSWER

Four zones, each repairing a failure mode of the last — a controlled evidence pipeline, not one model.



03 Results

PROOF FROZEN 244-QUESTION RUN



- Retrieval ceiling: R@10 = 0.258
 - French under-served
 - Abstention precision 0.654
 - Metrics are proxies
 - Corpus-tuned embedder
 - French ↔ Arabic retrieval
 - Graph-retrieval redesign
 - Expert legal review
- △ LIMITS
- 🕒 NEXT